1 Bridging the physical and digital worlds

This chapter covers

- Defining what a digital twin is
- Different levels of digital twin maturity
- What digital twins are good for
- How digital twins are used across industries
- Considerations when embarking on building a digital twin

Digital twins are virtual models of physical systems that continuously synchronize with real-world data to monitor current conditions and predict future behavior. NASA pioneered this approach decades ago with simulators for the Apollo program. Yet for years, digital twins remained accessible only to large, well-funded organizations that could afford the scarce, expensive sensors, computing capacity, and advanced analytics required to build them. That’s all changed.

Today, digital twins can be built by almost anybody, with an array of low-cost sensors, high-performance, pay-as-you-go computing, and powerful artificial intelligence and machine learning tools widely available.

Some think of digital twins as detailed 3D models. while others see them as glorified dashboards or traditional simulation systems. In this book, we define a digital twin as a system that combines elements from all these interpretations, but is not restricted to any single technology.

What exactly counts as a digital twin? Let’s start with a clear definition before exploring why they’ve become so popular.